pyCHM derivations

First-principles group theory of the Minimal and Next-to-Minimal Composite Higgs

Generated for the `pyCHM` library

June 21, 2026

Abstract

This note derives, from group theory alone, every quantity that `pyCHM` treats as *derived* (as opposed to model input, or numerically anchored to the independent `pypngb` engine). Each step is tied to the function that implements it and the test that verifies it in closed form, so the document and the code are one object: if a derivation here is wrong, a named test fails. The reference physics is Murnane, *The Landscape of Composite Higgs Models* (arXiv:2606.18364); conventions match the library exactly. Nothing here assumes the thesis is correct — the closed forms are recomputed from the generators, and where the thesis is the only source (the App. A7 form factors) it is cross-checked against the `pypngb`-anchored eigenvalue route (§5).

Contents

1 The CCWZ framework

A composite Higgs is a set of pseudo-Nambu–Goldstone bosons (pNGBs) of a spontaneously broken global symmetry $G \to H$. Write the generators of $\mathfrak{g} = \mathrm{Lie}(G)$ as the unbroken $T^a \in \mathfrak{h}$ and the broken $X^{\hat{a}} \in \mathfrak{g}/\mathfrak{h}$. The pNGBs $\pi^{\hat{a}}(x)$ parameterise the coset through the Goldstone matrix

$$U(\pi) \;=\; \exp\!\Big(\tfrac{i}{f}\,\pi^{\hat{a}} X^{\hat{a}}\Big), \tag{1}$$

with f the decay constant. Fermions are embedded in (generally incomplete) representations R of G; the Higgs dependence of every elementary–composite mixing is the matrix element of U in R,

$$\big\langle c \,\big|\, U_R(\pi) \,\big|\, E \big\rangle, \tag{2}$$

between the embedding E of the elementary field and a composite state c. The central claim of the library is that *these matrix elements are the only source of the s_h-dependence*, and that they are computable in closed form from the generators — never fitted. The rest of this document carries out that computation for SO(5)/SO(4) (MCHM) and SO(6)/SO(5) (NMCHM).

2 Minimal Composite Higgs: SO(5)/SO(4)

2.1 Generators and conventions

SO(5) acts on indices $0, \ldots, 4$. The unbroken SO(4) acts on $0, \ldots, 3$; the coset direction is index 4. In the vector (the $\mathbf{5}$) the Hermitian generators are

$$\big(T^{AB}\big)_{CD} \;=\; -\,i\big(\delta_{AC}\delta_{BD} - \delta_{AD}\delta_{BC}\big), \qquad 0 \le A < B \le 4. \tag{3}$$

The six T^{ab} ($a < b \le 3$) generate SO(4); the four broken generators are $X^{\hat{a}} = T^{\hat{a}4}$. With the Higgs vacuum expectation value along the $\hat{a} = 3$ direction, the only field that survives in unitary gauge is the physical Higgs h, and U becomes a single planar rotation.

▶ **Implemented in:** `groups.so5.gen_vector`, `groups/so5.py` **Verified by:** `test_so4_unbroken_generators_close_t`

2.2 The vector Goldstone matrix

Exponentiating the broken generator $X^3 = T^{34}$ with angle $\theta = h/f$ gives a rotation in the $(3,4)$ plane,

$$U_{\mathbf{5}}(\theta) = \exp\!\big(i\theta\, T^{34}\big) = \begin{pmatrix} \mathbb{1}_3 & & \\ & c_h & s_h \\ & -s_h & c_h \end{pmatrix}, \qquad s_h \equiv \sin\theta, \quad c_h \equiv \cos\theta = \sqrt{1 - s_h^2}. \tag{4}$$

The library stores everything in the variable $s_h = \sin(h/f)$ with $c_h = \sqrt{1 - s_h^2}$, avoiding an arcsin round-trip (machine precision). Eq. (4) is the thesis “explicit Goldstone matrix.”

▶ **Implemented in:** `groups.so5.U_vector`, `so5.U_vector_sym` **Verified by:** `test_U_vector_matches_thesis_goldst`

2.3 Lifting to tensor representations

A rank-k SO(5) tensor irrep is carried by the totally symmetric-traceless or totally antisymmetric part of $\mathbf{5}^{\otimes k}$. The Goldstone acts on each index,

$$\big(U_R\big)_{ba} = \big\langle E_b \,\big|\, U_{\mathbf{5}}^{\otimes k}\, E_a \big\rangle = \mathrm{Tr}\big(E_b^{\dagger}\, U E_a U^{\top}\big) \quad \text{(rank 2)}, \tag{5}$$

in an orthonormal basis $\{E_a\}$ of the irrep. The first three rungs are the **5** (rank 1), the **10** (antisymmetric rank 2) and the **14** (symmetric-traceless rank 2). The construction is dimension-agnostic, so the same code produces the **30** (rank-3 symmetric) and, with $n = 6$, the SO(6) tower of §3.

▶ **Implemented in:** `tensors.tensor_basis`, `tensors.U_rep_tensor`, `groups.so5.U_rep` **Verified by:** `test_tensors.py (dimension, unitarity, homomorphism)`

2.4 $\mathrm{SO}(4) = \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ branching

The unbroken SO(4) on indices $0, \ldots, 3$ is $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ via the 't Hooft combinations

$$J_i^{L/R} \;=\; \tfrac{1}{2}\Big(\tfrac{1}{2}\,\epsilon_{ijk} T^{jk} \;\pm\; T^{0i}\Big), \qquad i = 1, 2, 3. \tag{6}$$

Diagonalising the two Casimirs $C_L = \sum_i (J_i^L)^2$, $C_R = \sum_i (J_i^R)^2$ on the irrep basis (they commute) labels each sub-multiplet by (j_L, j_R) with $C = j(j+1)$. The branchings are

$$\mathbf{5} = \big(\tfrac{1}{2}, \tfrac{1}{2}\big) \oplus (0,0), \tag{7}$$
$$\mathbf{10} = \big(\tfrac{1}{2}, \tfrac{1}{2}\big) \oplus (1,0) \oplus (0,1), \tag{8}$$
$$\mathbf{14} = (1,1) \oplus \big(\tfrac{1}{2}, \tfrac{1}{2}\big) \oplus (0,0). \tag{9}$$

▶ **Implemented in:** `decompose.so4_decompose`, `decompose._su2_casimirs` **Verified by:** `test_branchings_match_th`

2.5 Dressings are matrix elements (derive, don't fit)

A claimed closed form $f(\theta)$ for an elementary–composite mixing is admissible only if it lies in the span of the single-basis overlaps $\{\langle E_b | U_R | E\rangle\}$. The library solves for the exact minimal-norm composite tensor c with $\langle c | U_R | E\rangle = f$ and *raises* if f is not realisable. This is the from-scratch guarantee: every dressing factor in the model files is *proven* to be a Goldstone matrix element, with symbolic residual exactly 0.

▶ **Implemented in:** `groups/derive.py::solve_composite` **Verified by:** `test_dressings_are_matrix_elements`

2.6 The 5-5-5 coefficients

Embed the up-component of q_L in the bidoublet, $E_{q_L} = (e_0 + e_3)/\sqrt{2}$, and t_R in the SO(4) singlet $E_{t_R} = e_4$. From (4), $U e_4 = s_h\, e_3 + c_h\, e_4$ and $U e_3 = c_h\, e_3 - s_h\, e_4$, $U e_0 = e_0$, so

$$\langle q_L | U_{\mathbf{5}} | t_R \rangle = \Big\langle \tfrac{e_0 + e_3}{\sqrt{2}} \,\Big|\, s_h\, e_3 + c_h\, e_4 \Big\rangle = \frac{s_h}{\sqrt{2}}, \qquad\qquad \big|\langle q_L | U | t_R\rangle\big|^2 = \frac{s_h^2}{2}, \tag{10}$$

$$\langle q_L | U_{\mathbf{5}} | q_L \rangle = \tfrac{1}{2}\big(1 + c_h\big) = \cos^2 \tfrac{h}{2f}. \tag{11}$$

These are exactly the $\Pi_L \sim s_h^2/2$, $\cos^2(h/2f)$ structures of the thesis 5-5-5 form factors, here *derived* from the overlaps.

▶ **Implemented in:** `so5.overlap_sym`, `mchm5.formfactor_pieces` **Verified by:** `test_coeffs_5_5_5`

2.7 The 14 channel weights and the $4/5$ enhancement

The t_R embeds in the SO(4) singlet of the **14**, the unit-normalised symmetric-traceless tensor

$$S = \frac{1}{\sqrt{20}}\,\mathrm{diag}(1,1,1,1,-4), \qquad \mathrm{Tr}S^2 = \frac{4+16}{20} = 1. \tag{12}$$

Dress it, $S \mapsto USU^\top$, and project onto the SO(4) channels of the **14**. The squared lengths in the singlet $(1,1)$, bidoublet $(\frac{1}{2},\frac{1}{2})$, and $(1,1)\to(3,3)$ channels are the closed trigonometric *channel weights*

$$W_{11} = \frac{(4c_h^2 - s_h^2)^2}{16}, \quad W_{22} = \frac{5}{2}\,s_h^2 c_h^2, \quad W_{33} = \frac{15}{16}\,s_h^4, \tag{13}$$

which satisfy unitarity exactly:

$$W_{11}+W_{22}+W_{33} = \left(1 - \tfrac{5}{2}s_h^2 + \tfrac{25}{16}s_h^4\right) + \tfrac{5}{2}s_h^2(1-s_h^2) + \tfrac{15}{16}s_h^4 = 1. \tag{14}$$

The thesis writes the t_R self-energy with an overall coupling $Y_T\sqrt{4/5}$. That normalisation is *not* an external input: it is the squared index-4 component of the embedding itself,

$$\boxed{\left|S_{44}\right|^2 = \left(\tfrac{-4}{\sqrt{20}}\right)^2 = \frac{16}{20} = \frac{4}{5}} \qquad \Longleftrightarrow \qquad \sqrt{\tfrac{4}{5}} = |S_{44}|. \tag{15}$$

Index 4 is the coset-singlet direction through which the top mass is generated; the **5**-singlet has weight 1 there (the unenhanced reference), so the 4/5 is a genuine representation-dependent enhancement. The thesis singlet-channel prefactor then follows as a *prediction*,

$$\frac{(4c_h^2 - s_h^2)^2}{20} \;=\; |S_{44}|^2\,W_{11} \;=\; \frac{4}{5}\cdot\frac{(4c_h^2 - s_h^2)^2}{16}, \tag{16}$$

which would fail for any inconsistent thesis number — the check is no longer a tautology that reads 4/5 off the prefactor 1/20.

▶ **Implemented in:** `decompose.channel_weights_sym`, `so5.embedding_sym` **Verified by:** `test_14_channel_weights_` `test_4_5_is_derived_from_the_embedding`

3 Next-to-Minimal Composite Higgs: SO(6)/SO(5)

3.1 Coset and broken generators

SO(6) acts on indices $0,\dots,5$. The unbroken SO(5) acts on $0,\dots,4$; the coset direction is index 5. The five broken generators split, under the $\mathrm{SO}(4)\subset\mathrm{SO}(5)$, into the Higgs bidoublet and a singlet:

$$X_B^a = T^{a5}\ (a=0,1,2,3) \quad \text{(the doublet } h_1,\dots,h_4), \qquad X_S = T^{45} \quad \text{(the singlet } s). \tag{17}$$

This is the thesis decomposition (its 6th index is our index 5). The 15 generators close the $\mathfrak{so}(6)$ algebra; the ten T^{ab} with $a<b\le 4$ are the unbroken SO(5).

▶ **Implemented in:** `so6.gen_vector`, `so6.UNBROKEN/BROKEN` **Verified by:** `test_so6_generators_close_the_algebra` `test_coset_split_10_plus_5`

3.2 The five-pNGB Goldstone matrix (closed form)

In unitary gauge the Higgs points along $\hat{a} = 3$ and the singlet along X_S, so the vector Goldstone is generated by

$$G = \theta_h\, L^{(3,5)} + \theta_s\, L^{(4,5)}, \qquad \theta_h = \tfrac{h}{f}, \quad \theta_s = \tfrac{s}{f}, \tag{18}$$

with $L^{(ab)}$ the real antisymmetric rotation generators. G is supported on the three indices $\{3, 4, 5\}$ and has rank 2, hence satisfies $G^3 = -\Theta^2 G$ with $\Theta = \sqrt{\theta_h^2 + \theta_s^2}$. Rodrigues' formula then gives the Goldstone in closed form,

$$U_6(\theta_h, \theta_s) = \mathbb{1} + \frac{\sin\Theta}{\Theta}\, G + \frac{1 - \cos\Theta}{\Theta^2}\, G^2. \tag{19}$$

Acting on the vacuum e_5 reproduces the thesis Goldstone field (eq. 474 there):

$$\Phi = U_6\, e_5 = \Big(0, 0, 0,\ \theta_h \tfrac{\sin\Theta}{\Theta},\ \theta_s \tfrac{\sin\Theta}{\Theta},\ \cos\Theta\Big)^{\top}, \tag{20}$$

i.e. $\Phi = \frac{1}{\Theta}\sin\Theta\,(h_1, \ldots, h_4, s, \Theta\cot\Theta)$ with $\Theta = \varphi/f$, $\varphi = \sqrt{\sum h_i^2 + s^2}$. Setting $\theta_s = 0$ returns the MCHM planar rotation (4) (with coset index $4 \to 5$).

▶ **Implemented in:** `so6.U6_vector`, `so6.U6_vector_sym`, `so6.goldstone_vacuum` **Verified by:** `test_goldstone_vac` `test_U6_vector_orthogonal_identity_homomorphism`

3.3 The representation tower

The fermion-partner irreps of SO(6) $\cong$ SU(4) are built and dressed by the same machinery: tensors via $n = 6$, spinors via the Clifford algebra. Their dimensions, SO(5) branchings (from the quadratic Casimir $C_5 = \sum_{a<b\le 4}(T^{ab})^2$), and SO(4) content are

irrep	construction	dim	SO(5) branching
$\mathbf{6}$	vector	6	$\mathbf{5} \oplus \mathbf{1}$
$\mathbf{15}$	antisym. rank 2 (adjoint)	15	$\mathbf{10} \oplus \mathbf{5}$
$\mathbf{20'}$	sym.-traceless rank 2	20	$\mathbf{14} \oplus \mathbf{5} \oplus \mathbf{1}$
$\mathbf{10}, \overline{\mathbf{10}}$	(anti)self-dual 3-form	10	$\mathbf{10}$ (each)
$\mathbf{4}, \overline{\mathbf{4}}$	Weyl spinor	4	$\mathbf{4}$

The Casimir calibration $C_5 \in \{0, \frac{5}{2}, 4, 6, 10\} \leftrightarrow \{\mathbf{1}, \mathbf{4}, \mathbf{5}, \mathbf{10}, \mathbf{14}\}$ identifies each sub-irrep with its multiplicity; the antisymmetric 3-form is 20-dimensional and splits, by the $\pm i$ eigenspaces of the Hodge dual $(\star T)_{ijk} = \frac{1}{3!}\epsilon_{ijklmn}T_{lmn}$, into the self-dual $\mathbf{10}$ and anti-self-dual $\overline{\mathbf{10}}$ (the SU(4) $\mathbf{10}+\overline{\mathbf{10}}$). The spinors come from six 8×8 Euclidean gammas $\Gamma^A = (\sigma_1 \otimes \gamma^a_{\mathrm{SO}(5)}, \sigma_2 \otimes \mathbb{1})$, split by chirality $\chi = \sigma_3 \otimes \mathbb{1}$; the $\mathbf{4}$ carries no $(\frac{1}{2}, \frac{1}{2})$ bidoublet, which is exactly the thesis reason the NM4DCHM uses the $\mathbf{6}$ (not the $\mathbf{4}$) for q_L.

▶ **Implemented in:** `so6.rep_basis`, `so6.U6_rep`, `so6.so5_content`, `so6_spinors.py` **Verified by:** `test_so6.py`, `test_so6_spinors.py`

3.4 Two-field dressing and the singlet pNGB

The **6** Goldstone dresses the elementary–composite overlaps with *both* angles. With $E_{q_L} = (e_0 + e_3)/\sqrt{2}$ and $E_{t_R} = e_5$, no embedding has support on index 4, so every overlap depends on θ_h, θ_s only through quantities even in θ_s; the potential is even in s, the vacuum sits at $\langle s \rangle = 0$, and the singlet mass is the s-curvature of the Coleman–Weinberg potential there. Because the q_L bidoublet dressing carries explicit θ_h-dependence distinct from the radial Θ the t_R feels, $V(h, s)$ is genuinely two-dimensional and the singlet acquires a finite, calculable mass — the defining NMCHM observable, absent in the MCHM. At $\theta_s = 0$ the mass matrices reduce *entry-for-entry* to the `pypngb`-anchored 5-5-5 model.

▶ **Implemented in:** `nmchm6.py (spec, assemble, potential, singlet_mass2)` **Verified by:** `test_nmchm6.py (reduction, even-in-`s`, singlet mass)`

4 The Coleman–Weinberg potential and observables

The one-loop pNGB potential is the supertraced kernel

$$V(s_h) = \sum_i \frac{c_i}{64\pi^2} m_i^2(s_h)^2 \log m_i^2(s_h), \qquad c_i = \{+3, +6, -12\} \tag{21}$$

for {neutral vector, charged vector, coloured Dirac}, with $m_i(s_h)$ the eigenvalues of the s_h-dependent mass matrices. Writing $V = -\gamma\, s_h^2 + \beta\, s_h^4$, the vacuum, Higgs mass and electroweak scale are the closed forms

$$\xi = \sin^2\frac{\langle h \rangle}{f} = \frac{\gamma}{2\beta}, \qquad m_h^2 = \frac{8\beta}{f^2}\,\xi(1-\xi), \qquad v_{\rm EW} = f\sqrt{\xi}, \tag{22}$$

and the Barbieri–Giudice tuning is $\Delta_{\rm BG} = \max_i |\partial \ln f / \partial \ln x_i|$ over the fundamental Lagrangian masses x_i. These relations are validated symbolically; the full mass matrices and the resulting $\xi, m_t, m_h, \Delta_{\rm BG}$ are anchored numerically to the independent `pypngb` engine to $< 0.1\%$.

▶ **Implemented in:** `routes.py, potential.py, spectrum.py, tuning.py` **Verified by:** `test_cw_kernel_and_coeffi`
`test_higgs_mass_scaling_closed_form, test_anchors.py`

5 The form-factor route and the oracle boundary

The thesis App. A7 form factors A_L, A_R, A_M, B are transcribed in `mchm5`; transcription proves faithful copying, not physical correctness. They are confronted with the `pypngb`-anchored eigenvalue route by comparing the top mass two ways. With the *custodial* parametrisation the form-factor route requires (a single q_L coupling L_q, i.e. $\Delta_{uL} = \Delta_{dL}$), the two coincide as $s_h \to 0$,

$$\frac{m_t^{\rm ff}(s_h)}{m_t^{\rm eig}(s_h)} = 1 - 0.755\, s_h^2 + \mathcal{O}(s_h^4) \xrightarrow[s_h \to 0]{} 0.999997, \tag{23}$$

the finite-s_h gap being the form-factor route's leading-order-in-s_h^2 truncation of the exact (non-polynomial) eigenvalue mass. The App. A7 form factors are therefore the correct leading-order two-point functions of the `pypngb`-anchored mass matrix — validated, not merely transcribed.

▶ **Implemented in:** `mchm5.formfactor_pieces, mchm5.fermion_mass` **Verified by:** `test_formfactor_route.py`

6 Beyond SO(N): the coset abstraction and SU(4)/Sp(4)

Everything above is built from one group-agnostic engine (`pychm.groups`): a coset G/H is a list of unbroken generators of H and broken generators X^a in some representation, and the Goldstone is $U(\pi) = \exp(i\,\pi^a X^a)$ – in unitary gauge a rank-≤ 2 rotation, so the closed Rodrigues form (19) serves every case. The MCHM and NMCHM are the instances SO(5)/SO(4) and SO(6)/SO(5); the *same code* builds $\mathrm{SU}(N)/\mathrm{Sp}(N)$ cosets, with the $\mathrm{SU}(N)$ Gell-Mann generators and the $\mathrm{USp}(N)$ subalgebra obtained from the Cartan involution $\theta(X) = \Omega\,\bar{X}\,\Omega^{-1}$ (Ω the symplectic form): $\mathfrak{usp}(N)$ is the $\theta = -1$ eigenspace, the coset the $\theta = +1$ eigenspace.

SU(4)/Sp(4) ≅ SO(6)/SO(5). The minimal pseudoreal Ferretti–Sannino coset SU(4)/Sp(4) has $15-10 = 5$ broken generators — the same five pNGBs (Higgs doublet + singlet) as SO(6)/SO(5). Under SU(4) ≅ Spin(6), Sp(4) ≅ Spin(5) the representations map as

$$\mathbf{4}_{\mathrm{SU}(4)} = \mathbf{4}^{\mathrm{SO}(6)}_{\mathrm{spinor}}, \quad \mathbf{6}^{[\mathrm{antisym}]}_{\mathrm{SU}(4)} = \mathbf{6}^{\mathrm{SO}(6)}_{\mathrm{vector}}, \quad \mathbf{15}_{\mathrm{SU}(4)} = \mathbf{15}^{\mathrm{SO}(6)}. \tag{24}$$

The library builds SU(4)/Sp(4) independently (its own generators, Goldstone, and the antisymmetric $\mathbf{6} = [\mathbf{4} \otimes \mathbf{4}]_A$) and cross-checks it against the existing SO(6) code, which is an *exact oracle*: the $\mathbf{4}$ branches to a single $\mathbf{4}$ of Sp(4) and the $\mathbf{6}$ to $\mathbf{1} \oplus \mathbf{5}$, matching the SO(6) spinor and vector; and the strong statement of the isomorphism — the $\mathbf{6}$ generators span the full 15-dimensional $\mathfrak{so}(6) \cong \mathfrak{su}(4)$ algebra and act irreducibly (commutant = scalars, by Schur) — holds, so the SU(4) antisymmetric $\mathbf{6}$ *is* the irreducible SO(6) vector. The new coset thus reproduces the NMCHM group theory while sharing all of its tooling.

▶ **Implemented in:** `groups/lie.py (su_generators, sp_subalgebra), groups/coset.py, groups/su4sp4.py`
Verified by: `test_su4sp4.py`

7 SU(5)/SO(5) (littlest Higgs) and the coset landscape

SU(5)/SO(5) is a type-AI symmetric space: the Cartan involution $\theta(X) = \bar{X}$ (complex conjugation) splits $\mathfrak{su}(5)$ into the unbroken $\mathfrak{so}(5)$ (10 imaginary-antisymmetric generators, the $\theta = -1$ eigenspace) and the coset (the $\theta = +1$ eigenspace, 14 real-symmetric generators). The same engine that built the SU(4)/Sp(4) split builds this one (`lie.cartan_split` with the conjugation involution). The 14 pNGBs are the symmetric-traceless $\mathbf{14}$ of SO(5), which branches under custodial SO(4) as

$$\mathbf{14} \;=\; (3,3) \oplus (2,2) \oplus (1,1) \;=\; \text{complex triplet } \phi \;\oplus\; \text{Higgs doublet } h \;\oplus\; \text{singlet } \eta, \tag{25}$$

the littlest-Higgs pNGB content. The Goldstone in the fundamental $\mathbf{5}$ is $U = \exp(i\,\theta^a X^a)$ over the coset generators (a complex unitary rotation, unlike the real SO(N) one); along the Higgs axis the $(2,2)$ overlap magnitude reduces to the MCHM value $|\langle q_L|U|t_R\rangle|^2 = s_h^2/2$, so the top and Higgs match the 5-5-5 there. The symmetric $\mathbf{15}$ of SU(5) (the fuller partner rep) is built with the SU(N) tensor basis (no δ-trace removal: `tensor_basis(…,group='SU')`) and branches $\mathbf{15} = \mathbf{14} \oplus \mathbf{1}$ under SO(5). The genuinely new observables are the singlet and triplet pNGB masses, the curvatures of the two-extra-field Coleman–Weinberg potential at the vacuum (even in η, ϕ, so $\langle\eta\rangle = \langle\phi\rangle = 0$).

▶ **Implemented in:** `groups/lie.py (cartan_split), groups/su5so5.py, su5so5.py` **Verified by:** `test_su5so5.py`

The landscape. A composite Higgs needs a custodial $\mathrm{SO}(4) \cong \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R \subset H$ and a $(2,2)$ Higgs in the coset G/H. Chala & Fonseca (arXiv:2309.10635) enumerate all compact G/H with ≤ 13 pNGBs satisfying this (642 models; complete up to 13 NGBs). `groups.landscape` reproduces a slice from pyCHM's own `Coset`: it branches each coset under the custodial SO(4) and flags the $(2,2)$, recovering MCHM $(2,2)$, NMCHM $(1,1)+(2,2)$, and littlest-Higgs $(1,1)+(2,2)+(3,3)$, and discovering SO(7)/SO(6), SO(8)/SO(7) (see `docs/COSET_LANDSCAPE.md`).

▶ **Implemented in:** `groups/landscape.py` **Verified by:** `test_landscape.py`

Provenance: derived vs. input vs. anchored

Derived (this document)
: the Goldstone matrices U, U_6; the rep lifts; the SO(4)/SO(5) branchings; the 5-5-5 coefficients; the **14** channel weights and the $4/5 = |S_{44}|^2$ enhancement; the full SO(6) tower. Each closes symbolically (a named test asserts `simplify`$(\cdot) = 0$).

Input (model structure)
: the partner content — which composite multiplets exist, the embeddings, the partner masses. Taken from the thesis/`pypngb`, not derived.

Anchored
: the full mass matrices and the physical observables, validated against `pypngb` ($< 0.1\%$). The NMCHM has no public engine and is anchored indirectly by its exact reduction to the 5-5-5 at $\langle s \rangle = 0$, with the new singlet observable cross-checked internally (stencil independence + parabolic fit).

After §2.7 and §5, no quantity that feeds an observable is a thesis read-off: correctness rests on `pypngb` + group theory.